

EDUCATION

Ph.D. in Machine Learning

Nov. 2024 – Curr.

National and Kapodistrian University of Athens, Dept. of Informatics and Telecommunications

- Supervisor: Christos Tzamos.
- Research interests: reasoning in Transformers, Combinatorics, and Neural Networks theory.
- Published at: **NeurIPS** (1), **ICLR** (1).

M.Sc. in Statistical Science

Oct. 2020 – Sep. 2021

University of Oxford, Dept. of Statistics, Lincoln College

- Thesis: *Authorship Attribution in Ancient Greek Texts*, co-supervised by Yannis Assael, Yorgos Deligiannidis, and Thea Sommerschild.
- Grade: **Merit**.

B.Sc. in Mathematics

Sep. 2016 – Aug. 2020

University of Patras, Dept. of Mathematics

- Thesis: *Mathematical Techniques in Data Science and Machine Learning*, under the supervision of Michael N. Vrahatis – Grade: **10.00/10.00**.
- Grade: **8.90/10.00** (Distinction), equivalent to first-class honors (**top 1%**).

EXPERIENCE

Ph.D. Researcher

Jul. 2024 – Curr.

Archimedes, Athena Research Center

- Conducting theory-based research on reasoning in Transformer models and Neural Networks theory.
- Supervisor: Christos Tzamos.

Visiting Ph.D. Researcher

Mar. 2026 – May 2026

Massachusetts Institute of Technology, Computer Science and AI Laboratory

- Investigating the properties of reliability and robustness in generative models.
- Supervisor: Costis Daskalakis.

Research Assistant

Apr. 2024 – Jun. 2024

Technical University of Denmark, Dept. of Applied Mathematics and Computer Science

- Explored Deep Learning theory via Riemannian geometry.
- Supervisor: Georgios Arvanitidis.

Research Assistant

Apr. 2023 – Dec. 2023

Czech Technical University in Prague, Faculty of Electrical Engineering

- Focused on the Mini-batch Stochastic Subgradient Projection method.
- Supervisors: Vyacheslav Kungurtsev, and Jakub Mareček.

Research Assistant

Apr. 2022 – Apr. 2024

Athena Research Center, Institute for the Management of Information Systems

- Researched the interface of NLP, Privacy, and Security tasks using LLMs.
- Supervisor: Constantinos Patsakis.

Research Assistant

Oct. 2021 – Mar. 2022

Max Planck Institute for Mathematics in the Sciences

- Worked on the authorship attribution task with a focus on Plutarch's texts using LLMs.
- Supervisors: Ivan P. Yamshchikov, and Jürgen Jost.

PUBLICATIONS

C: Conference. J: Journal. †: equal contribution.

- C_3 A Derandomization Framework for Structure Discovery: Applications in Neural Networks and Beyond.
Nikos Tsikouras, Yorgos Pantis, Ioannis Mitliagkas, Christos Tzamos. **ICLR 2026**.
- $C_2^\dagger$ Teaching Transformers to Solve Combinatorial Problems through Efficient Trial & Error.
Panagiotis Giannoulis, Yorgos Pantis, Christos Tzamos. **NeurIPS 2025**.

- J₁ The Anatomy of Deception: Measuring Technical and Human Factors of a Large-scale Phishing Campaign.*
Anargyros Chrysanthou, Yorgos Pantis, Constantinos Patsakis. **Computers & Security 2024.**
- C₁ BERT in Plutarch's Shadows.*
Ivan P. Yamshchikov, Alexey Tikhonov, Yorgos Pantis, Charlotte Schubert and Jürgen Jost. **EMNLP 2022.**

HONORS & AWARDS

4-year Full Scholarship

Jul. 2024 – Curr.

Archimedes, Athena Research Center

Awarded to fund my doctoral studies based on my academic background.

Research Infrastructure Grant

Dec. 2024 – May 2026

GRNET – Greek Research & Technology Network

Provided access to high-performance computing systems for my doctoral research experiments.

Full Annual Scholarship

Oct. 2020 – Aug. 2021

Bodossaki Foundation

Granted to support my master's studies in recognition of my undergraduate academic performance.

Covid-19 Assistance Fund

Jun. 2021

University of Oxford, Lincoln College

Financial assistance awarded during my master's studies amid the pandemic.

Vivian Green Student Assistance Scholarship

Feb. 2021

University of Oxford, Lincoln College

Conferred to support the advancement of my master's studies.

4-year Annual Scholarship

Sep. 2016 – Sep. 2020

Greek State Scholarships Foundation

Awarded throughout my bachelor's studies in recognition of sustained academic excellence.

4-year Consistently in the Top 1%

Sep. 2016 – Sep. 2020

University of Patras, Dept. of Mathematics

Ranked among the top 1% of students for the entire duration of my bachelor's program.

Ranked Second Among Admitted Students

Sep. 2016

University of Patras, Dept. of Mathematics

Placed second among admitted candidates to the bachelor's degree program.

TALKS & SERVICE

Selected Talks

Transformers Reasoning

- Harvard University May 2026
- Northeastern University May 2026
- Boston University Apr 2026
- MIT Apr 2026
- HSBC Sep 2025
- Archimedes, Athena Research Center May 2025
- AI & Education Outreach*
- Greek Institute of Education Policy Jan 2026

Reviewer

- NeurIPS, ICLR, ICML, TMLR. 2025 – Curr.

Teaching Assistant

National and Kapodistrian University of Athens, Dept. of Informatics and Telecommunications

- *Computational Learning Theory.* Fall Sem. 2025
- *Artificial Intelligence II (Deep Learning for NLP).* Fall Sem. 2025

Volunteer

- *Archimedes, Athena Research Center* Sep. 2025 – Curr.
- Organized Greece's AI competition; trained students for the International and European AI Olympiad.